Case Study Of AREN'S FANCY WORLD

Mid-Term Report for Business Data Management Capstone Project

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1 Contents

1. EXECUTIVE SUMMARY: -

This midterm report focuses on the business operations of *Aren's Fancy World*, a local stationery and gift store located in Ghaziabad. The primary challenges faced by the business include **inventory spoilage due to overstocking**, **customer retention issues**, and the **lack of a structured inventory management system**.

To understand and address these issues, primary sales data was collected for 3 months, covering over **490 transactions**. The dataset includes variables such as **product type**, **category**, **unit sold**, **selling price**, **cost price**, and **date of sale**. Preliminary descriptive analysis revealed that:

- Categories like **Gifts** and **Stationery** had the highest frequency of sales.
- Certain products showed **zero or very low movement**, indicating poor shelf turnover.
- **Profit margins** varied significantly across products, with some yielding negative margins.
- A noticeable **seasonal trend** was visible, with spikes in certain weeks (likely influenced by festivals or school cycles).

The methodological approach includes **data cleaning**, **categorical grouping**, and **statistical summarization**, followed by visualizations to explore patterns. The upcoming sections detail:

- A structured **metadata table** with variable descriptions.
- **Descriptive statistics** including mean, median, and standard deviation for numerical variables.
- An explanation of the **data collection and preprocessing methods** used.
- Initial **visual findings**, focusing on product-level performance and potential waste risks.

This report lays the groundwork for the final phase, where we will use these insights to recommend **data-backed improvements in stock planning and customer engagement strategies**.

2. PROOF OF ORIGINALITY: -

To verify the authenticity of the data, the supporting evidence is provided below:

- **Business Name:** Aren's Fancy World
- **Owner's Name:** Mr. Amit Aren

- **Business Address**: Orbit Plaza SHOP NUMBER UG 25 GROUND FLOOR ORBIT PLAZA, near AXIS BANK ATM, Crossings Republik, Ghaziabad, Uttar Pradesh 201016

All data and files are available in the Google Drive Folder: FOLDER

Video of Interaction: Interaction with Mr. Amit Aren. This video can be accessed through the following Google Drive link: Video

Letter from the Organisation: Stamped and signed letter from the organisation. Can be accessed through the Google drive link: Letter

Images of the Organisation: These images are included in Appendix A. They can be accessed through the following Google Drive link: Images Folder

3. METADATA: -

- **Data Format:** CSV and Excel (.xlsx file format)
- **Data Range:** November 2024
- **Data collection:** Data was collected through the bill book and the inventory was created by merging all the unsold products on a daily basis.
- **Data used can be accessed through the following links:**
  - Inventory data: Inventory Data
  - Monthly sales data: Sales Data
  - Daily sales data: Daily Sales
  - Monthly Purchase data: Purchase Data
- **Data Description:**
  The dataset consists of information about stationery items including:
  - Item names and categories
  - Quantity purchased and sold
  - Purchase cost and revenue generated
  - Remaining inventory at the end of the month

The owner willingly gave me the access to the sales, purchase and inventory data.

1. **Sales data:** The owner shared an excel sheet comprising of the sales report containing details of specific item names, quantities sold, and the total revenue generated per item only for the stationary items sold by the shop.
2. **Purchase data:** The owner shared an excel sheet comprising of the purchase report containing details of specific item names, units purchased, and the total cost per item only for the stationary items sold by the shop.
3. **Inventory data:** The owner shared an excel sheet comprising of the items in the inventory along with the number of units of the stationary items left after the month.

Variables Overview: -

Variable Name	Description	Unit	Data Type
Item	Name of stationary item	-	Categorical
Units	Number of units sold, purchased or present in inventory for each item	-	Numerical
Price (per unit)	The price of a unit of each item during owner's purchase	₹	Numerical
Overall Cost	Overall amount paid by the oner for purchasing the items	₹	Numerical
Sales Price (per unit)	The sales price of a unit of each item	₹	Numerical
Overall Revenue	The revenue generated by the sales of n units of an item	₹	Numerical

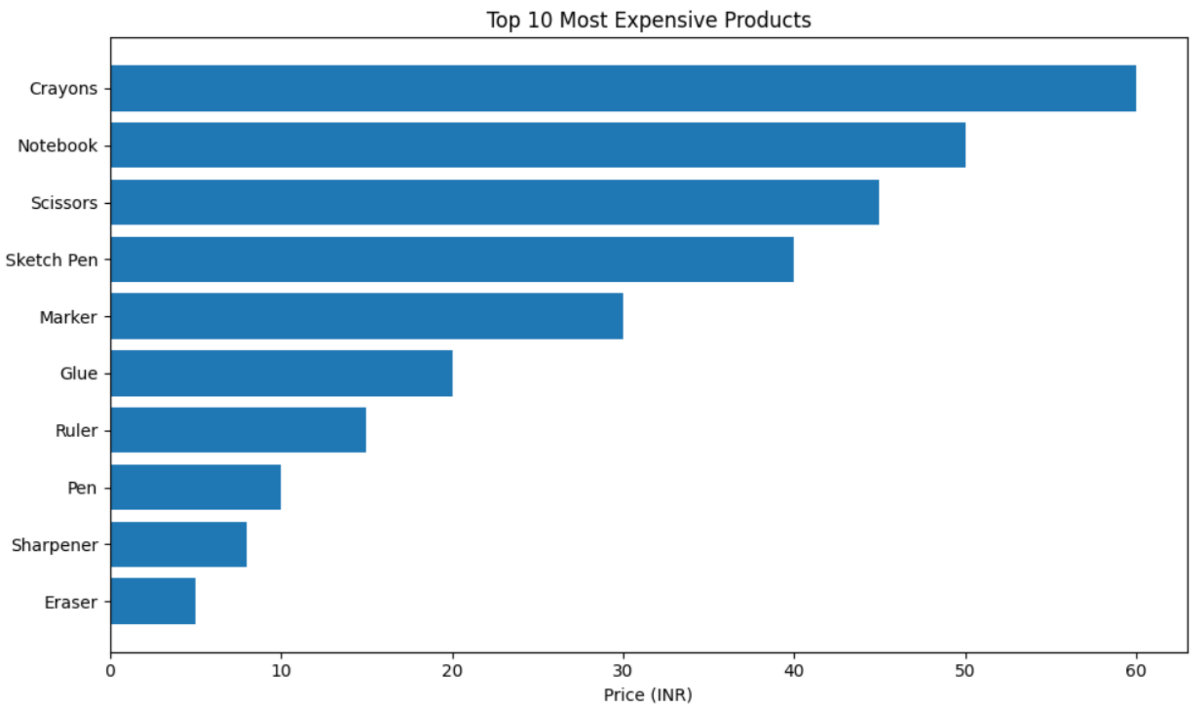
4. DESCRIPTIVE STATISTICS: -

The use of CSV and Excel files made performing descriptive statistics over collected data much easier. I have condensed the information for descriptive statistics below relevant for both sales and inventory data.

4.1 Inventory Data: -

The Inventory data was analysed using python in Google Colab. The source code can be seen here: INVENTORY Analysis

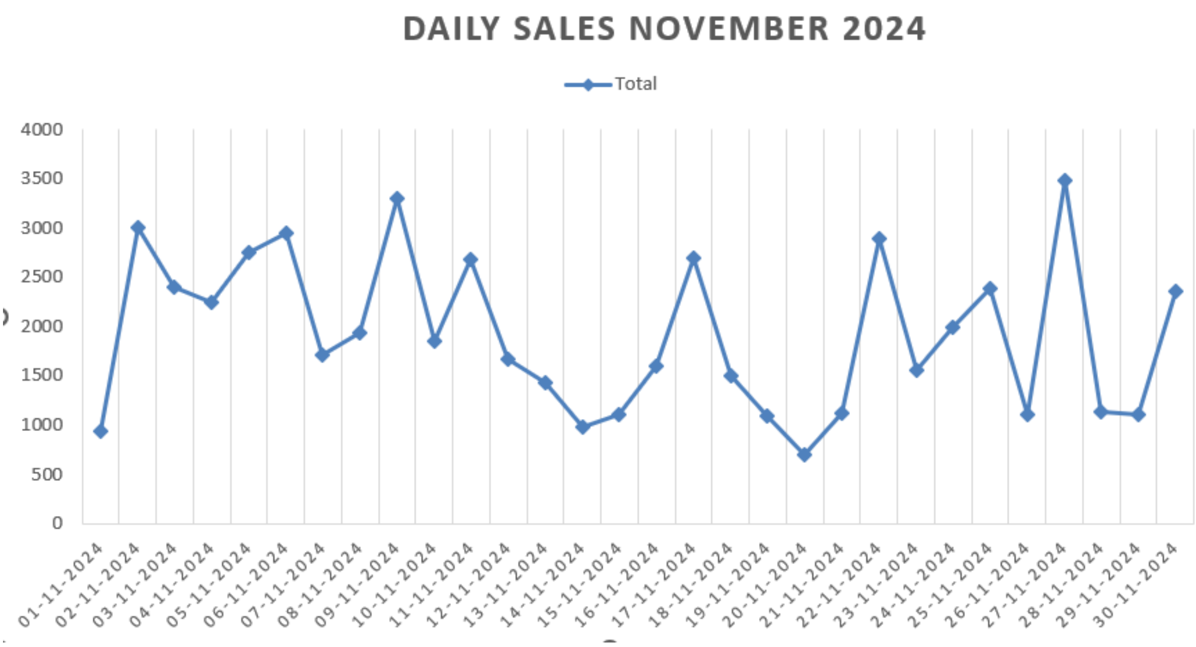
DESCRIPTIVE STATISTICS	AMOUNT(₹)
SUM	₹4588
MEAN	₹101.95
MEDIAN	₹28
MINIMUM PRICE OF PRODUCT	₹2
MAXIMUM PRICE OF PRODUCT	₹1215
STANDARD DEVIATION	₹227.291
TOTAL UNIQUE PRODUCTS	46
TOTAL BRANDS	13



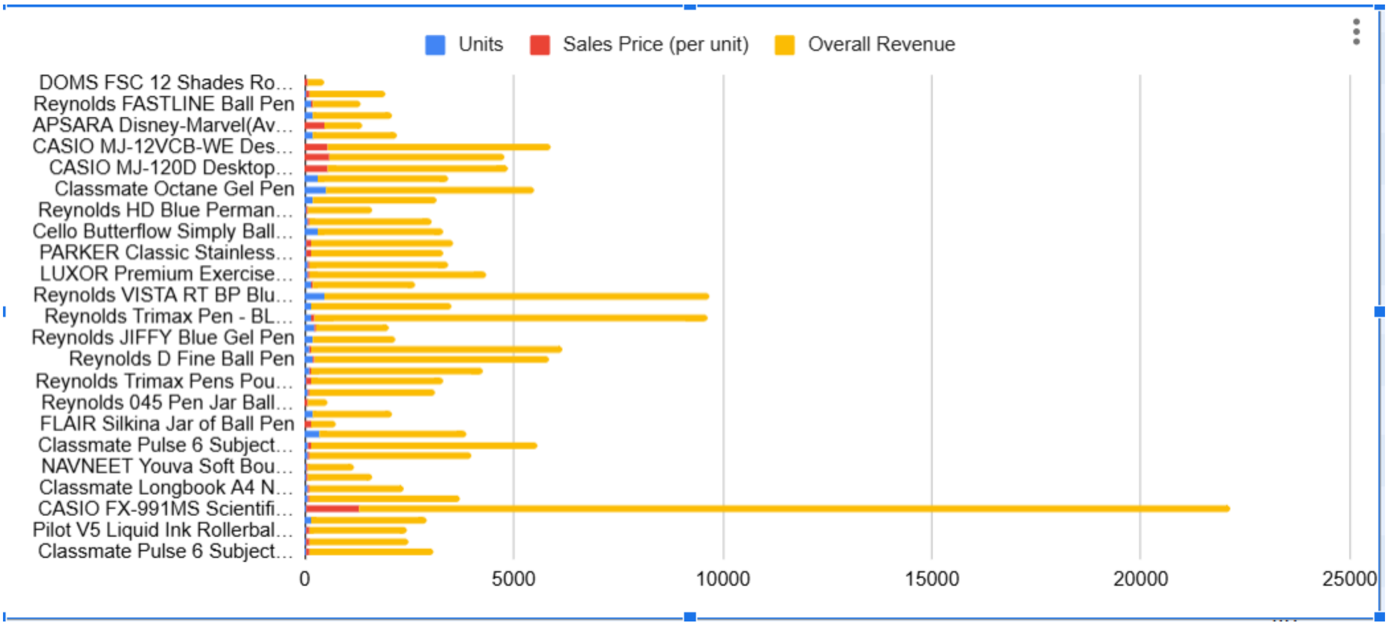
4.2 Sales Data: -

Measure	Definition
Sum	Total items sold
Mean	Average items sold
Median	Middle value
Minimum	Least revenue item
Maximum	Maximum revenue item
Standard Deviation	The measure of spread of values from mean in single sample.

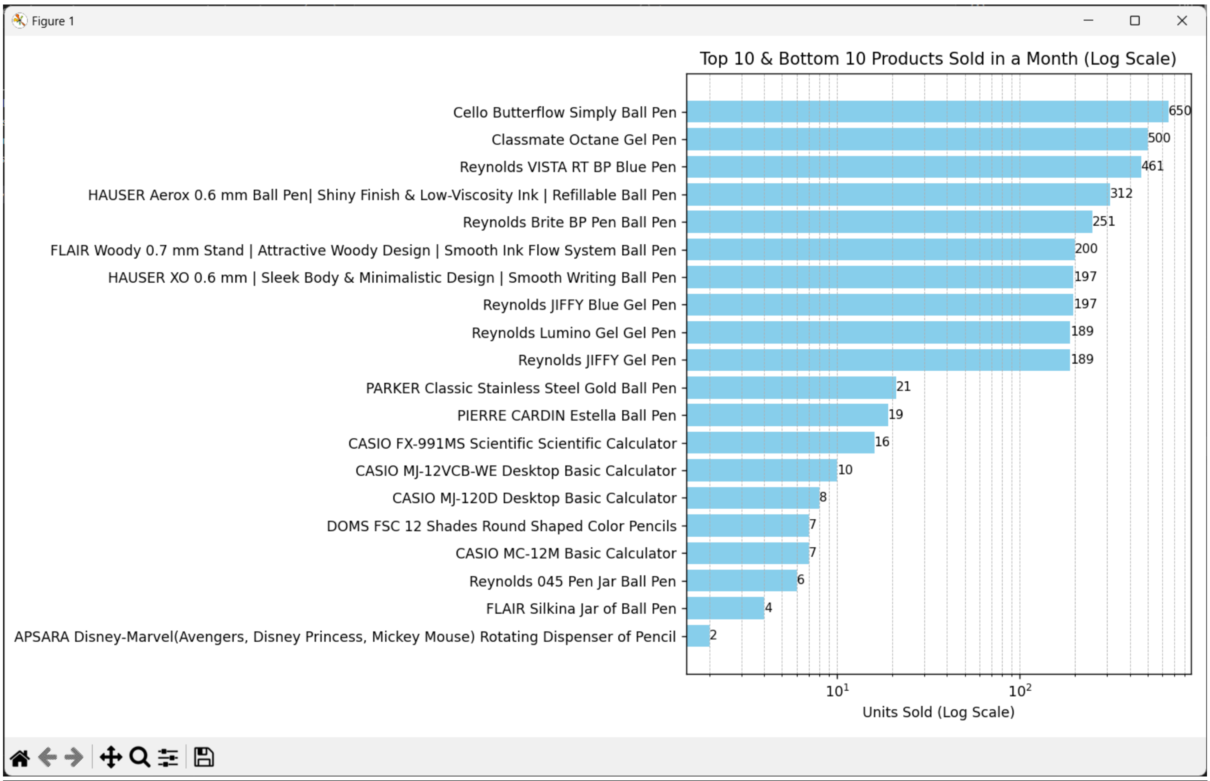
Measure	Quantity	Total Revenue
Sum	5244	160279
Mean	116.533	3561.755
Median	68	3000
Minimum	2	406
Maximum	500	20800
Standard Deviation	120.883	3250.255



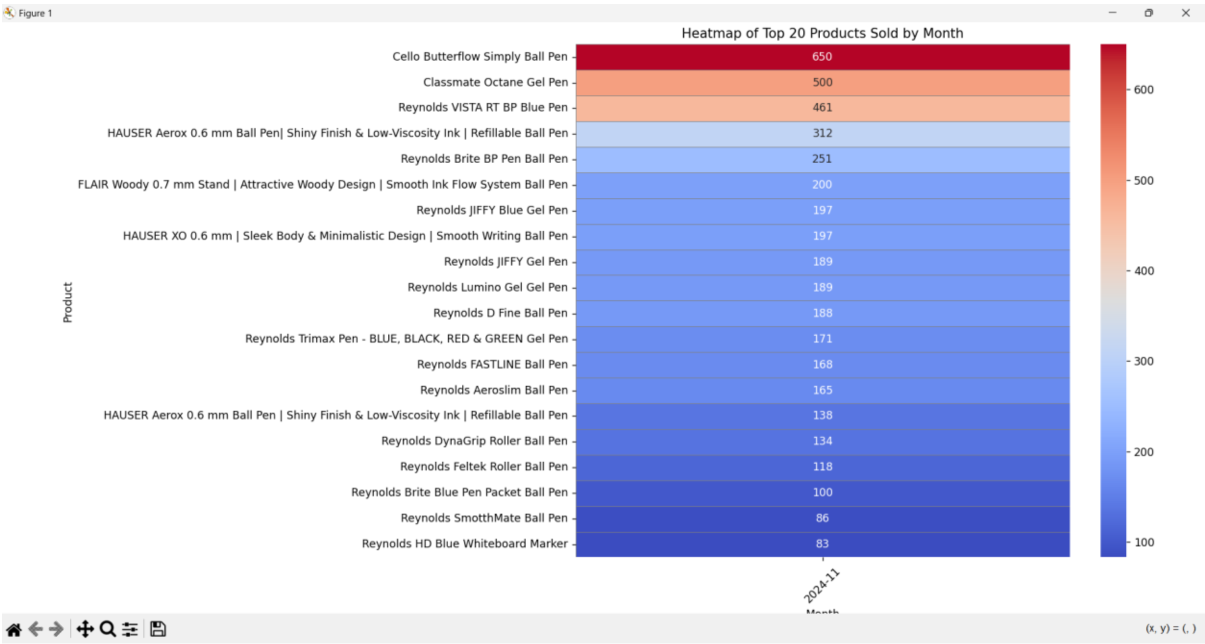
Item-wise revenue generated



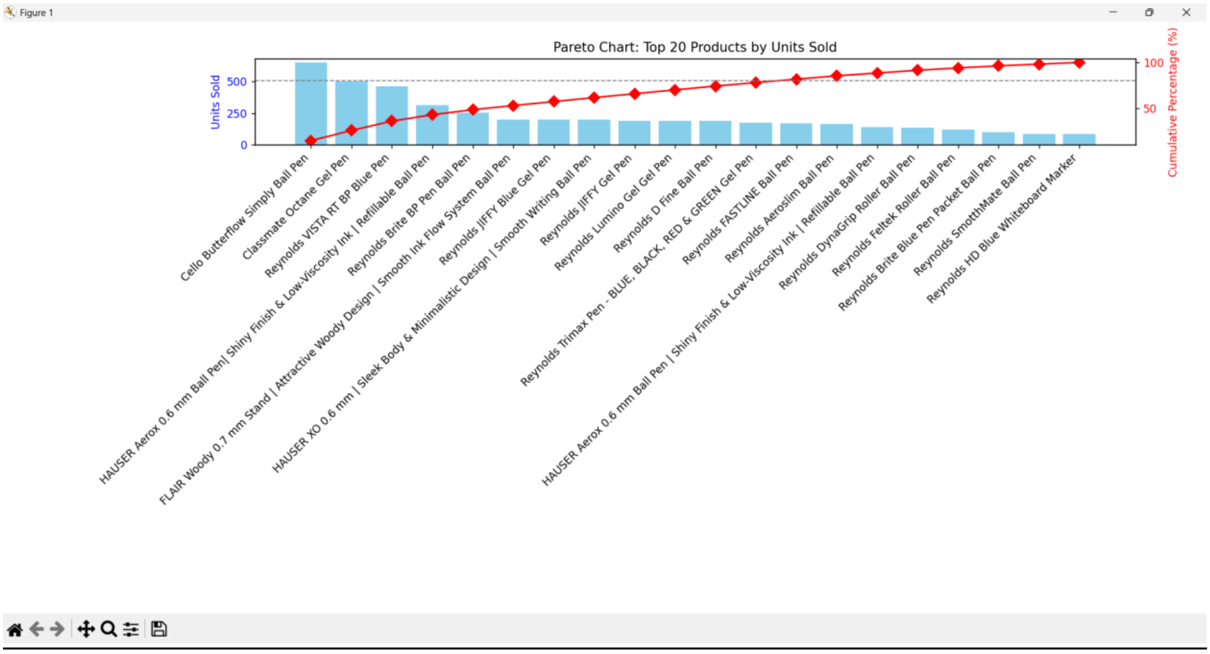
Top 10 & Bottom 10 Products sold in a month (using log scale)



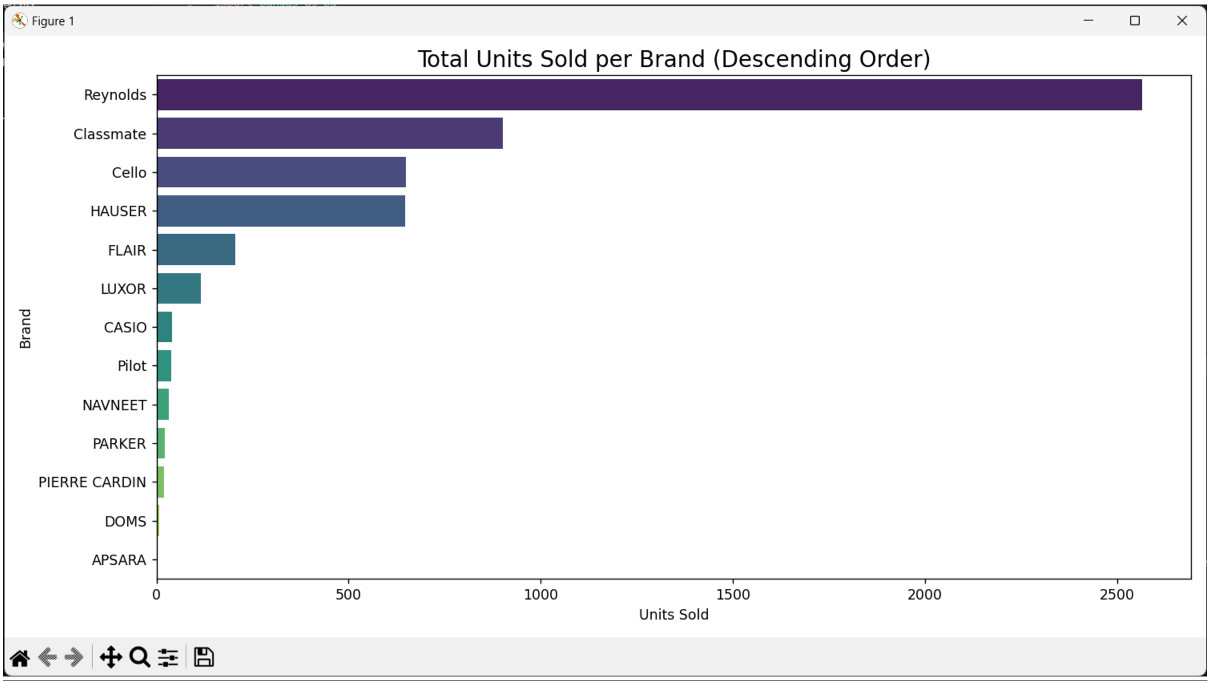
Heatmap of top 20 products sold



Pareto Chart for top 20 products sold



Brand-wise monthly sales bar chart



5. Detailed Explanation of Analysis Process & Methods: -

The analysis process for the project involves accumulation of data and then using tools like Python and Excel, and concepts from BDM theory to figure out business problems. My main motive behind this was that the results I work towards should be meaningful to me and the owner, and not just devolve to senseless graphs, though I still tried to incorporate and learn new concepts.

Analysis is divided into 2 parts, as follows:

1. General monthly trends
2. Trends which will help us tackle problem 3 (mentioned in the proposal document) i.e. Inventory Management issues

1. **Data Collection & Understanding the Business:** The analysis began with collecting past transaction records, sales data, and inventory details from Aren’s Fancy World, a stationery and gift shop. A short interview with the owner provided insights into key challenges, such as inventory management, customer retention, and handling special order requests that often lead to surplus stock.

2. **Software used:** Next onto the analysis part, once we had the data in excel format it was really easy to analyse it using BDM theory and data analysis skills learned.
   I primarily used Python and a few handy functions in excel like SUM, UNIQUE etc. The choice was simple because of extensive libraries present in python like pandas and matplotlib. Pandas helped me convert the data into dataframes that can be easily manipulated like we do with excel sheets but this is faster as you don’t have to manually do it. Matplotlib is the main python library for plotting graphs which helped me visualise the graphs and then present them in my analysis.

3. **Time-Series Analysis:** This approach is particularly effective for retail and inventory data, which is inherently time-dependent. By analyzing daily sales trends, brand performance, and product category patterns over time, we gain valuable insights into the store’s operational efficiency and financial health. This method is especially useful in identifying sales cycles, customer demand fluctuations, and the impact of specific events (like holidays). For instance, it helps us monitor underperforming brands such as APSARA, assess peak sales periods, and refine inventory and pricing strategies to maximize profitability and customer satisfaction.

4. **Pareto Chart Analysis:** We used a Pareto chart on the top 20 products to identify which items contributed most to revenue. Even within this group, a few products dominated sales, reflecting the 80/20 rule. This helped us focus on high-impact items for stocking and promotions, while also showing that the APSARA brand didn’t appear among top performers—highlighting a need for strategic review.

5. **<u>Communication:</u>** Interestingly, communication with the shop owner and workers was one of the bottle-neck in my project as I was not able to open up with the shop owner and it was difficult to convince them with radical new ideas. It was a bit easier once I talked to the owner's son who was around my age and understood the value of technology much more.

6. **<u>Quantitative Analysis Methods:</u>** Spreadsheets & Excel Tools: Used for calculating total credit across customers, analysing sales trends, and computing costs. Functions like SUM, AVERAGE, Median, Minimum, Maximum, Standard Deviation and Charts.

7. **<u>Qualitative Insights from Business Owner:</u>** Conversations with the owner revealed additional concerns, including customer returns, difficulty in selling bulk-ordered items, and competition from online platforms. Possible solutions included introducing new product categories, exploring online sales channels, and implementing stricter return policies.

8. **<u>Key Findings & Recommendations:</u>**
   - Improve inventory forecasting using past sales trends.
   - Introduce return policies to minimize loss from customer returns.
   - Explore online selling platforms for better market reach.
   - Implement bulk order tracking to ensure demand before stocking special items.

6. Results and Findings: -

<u>The key observations on the net sales and the net revenue:</u>

A consistent pattern can be seen between the sales and the cost of items.

6.1 General Monthly Trends: -

6.1.1 Total Daily Sales:

The daily sales trend throughout November displayed frequent fluctuations, with several pronounced peaks and troughs. Key spikes occurred around the 11th, 18th, and 28th of the month—each day exceeding ₹3000 in revenue—suggesting periodic promotional boosts or seasonal demand surges. Lows were observed around the 1st, 15th, and 21st, with revenue dipping close to ₹1000 or below, potentially pointing to stockouts, low footfall, or operational downtimes.

6.1.2 Daily Sales by Product Category:

6.2 Inventory Management Insights: -

6.2.1 Top 10 & Bottom 10 Products Sold:

The log-scale bar chart highlights a vast sales disparity between high- and low-performing products. The Cello Butterflow Simply Ball Pen, Classmate Octane Gel Pen, and Reynolds VISTA RT BP dominate the top-sellers list with 461–650 units sold. In stark contrast, bottom-tier products like APSARA Disney Pencil Dispenser and FLAIR Silikna Jar moved fewer than 5 units. Such poor-performing SKUs could be indicative of either limited appeal or shelf visibility and may require a review of inventory or marketing efforts.

6.2.2 Heatmap of Top 20 Products Sold:

The heatmap visualization of the top 20 products offers insights into sales consistency. Items like Cello Butterflow, Classmate Octane, and Reynolds VISTA stand out with dense, warm colors—indicating strong and steady movement throughout the month. Conversely, products like Reynolds SmoothMate and HD Blue Whiteboard Marker display lighter shades, hinting at sporadic or late-month sales activity. These trends can help identify products with sustained demand versus those with erratic sales patterns.

6.2.3 Pareto Chart for Top 20 Products Sold:

The Pareto analysis clearly validates the 80/20 principle, where just a handful of items contribute to the majority of sales. The top 5 items alone account for more than 60% of total unit sales. This indicates that a focused inventory strategy—prioritizing these high-movement items—could significantly improve sales efficiency. Products on the right end of the chart offer marginal returns and could be streamlined or bundled to improve throughput.

6.2.4 Brand-wise Monthly Sales:

The horizontal bar chart depicting brand-wise sales showcases Reynolds as the clear leader, surpassing 2500 units sold. Classmate, Cello, and HAUSER follow but with a noticeable gap. Mid-tier performers like FLAIR and LUXOR still maintain relevance, whereas brands such as DOMS, APSARA, and PIERRE CARDIN lag significantly. This brand distribution reveals an opportunity to capitalize on strong-performing brands while reconsidering the value of underperformers in the product lineup.

The statistical analysis of November 2024 sales revealed several actionable insights that could help the store improve performance and address current challenges. One key takeaway is the disproportionate contribution of a few top-performing products—particularly writing instruments like the Cello Butterflow, Classmate Octane, and Reynolds VISTA pens—to overall sales volume. These items consistently outperformed others, as seen across the bar charts, heatmap, and Pareto analysis.

A strategic solution that emerged during the analysis is the potential to introduce a private-label product line for these high-demand items, especially in the ball pen and gel pen categories. Given their daily utility and consistent sales, branding and selling such staples under the shop's own label could increase margins and brand recognition. Additionally, reviewing and optimizing shelf space for underperforming brands and products could further streamline inventory and reduce waste.

7. Appendix